

Linear Inequality:

Qno 1)

$$-2(m-3) < 5(m+1) - 12$$

$$-2m + 6 < 5m + 5 - 12$$

$$-2m - 5m < 5 - 12 - 6$$

$$-7m < +13$$

$$m < \frac{13}{7}$$

Qno 3)

$$-14 < -7(3x+2) < 1$$

$$-14 < -21x \quad -14 < 1$$

$$0 < -21x < 15+1$$

$$0 > x > \frac{15}{-21}$$

$$-\frac{5}{7} < x < 0$$

Qno 2)

$$2(1-x) + 5 \leq 3(2x-1)$$

$$2 - 2x + 5 \leq 6x - 3$$

$$-2x - 6x \leq -3 - 5 - 2$$

$$-18x \leq -10$$

$$x \leq \frac{5}{9}$$

Qno 4)

$$-3 < \frac{3}{2}(2-x) \leq 5$$

$$-3 < 3 - \frac{3x}{2} \leq 5$$

$$-3-3 < -\frac{3x}{2} \leq 5-3$$

$$-6 < -\frac{3x}{2} \leq 2$$

$$-6 \times \frac{2}{3} > -\frac{3x}{2} \times \frac{2}{3} \geq 2 \times \frac{2}{3}$$

$$-4 > x \geq -\frac{4}{3}$$

$$-\frac{4}{3} \leq x < -4$$

Quadratic Inequalities

Qno 1)

$$2x^2 - 7x + 3 > 0$$

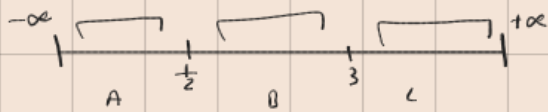
$$2x^2 - (1+1)x + 3 =$$

$$2x^2 - 6x - x + 3 > 0$$

$$2x(x-3) - 1(x-3) > 0$$

$$(2x-1)(x-3)$$

$$x = \frac{1}{2} \quad x = 3$$



A	$x < \frac{1}{2}$	$\frac{1}{2}$	+
B	$\frac{1}{2} < x < 3$	2	-
C	$x > 3$	4	+

$$x \in (-\infty, \frac{1}{2}) \cup (3, \infty)$$

